

SEAN KHOMPHENGCHAN

CURRENT ADDRESS
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SOCIALS
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OBJECTIVE

To obtain a fall internship or Full-Time position in a Guidance, Navigation, & Controls or Software related position applicable to the aerospace industry.

EDUCATION

Purdue University, West Lafayette, IN December 2023
Bachelor of Science in Aeronautical & Astronautical Engineering
Specializations: Dynamics and Control
Minor: Astronomy
GPA: 3.19 / 4.00
Relevant Coursework: Dynamics, Multi Agent Autonomy, Control Systems, and Satellite Navigation and Positioning
Technical Skills: MATLAB, Simulink, Python, C, C++, Siemens NX, Kalman Filtering, Git, Agile Development

TECHNICAL & RELEVANT EXPERIENCE

Simulation and Modeling Intern June 2022 – Present
Raytheon Missiles and Defense, Arlington VA

- Developed algorithms to predict and simulate solutions in systems and to conduct threat assessments
- Briefed executives all the way to the president of the company on work accomplished and impact of
- Directly contributed to a F-35 engine study that led to the winning of a \$5.2-billion contract for an engine upgrade
- Utilized Object Oriented Python and Java to build campaign level simulation software

Undergraduate Teaching Assistant for the First Year Engineering Program August 2021 – Present
Purdue University, West Lafayette IN

- Provide guidance to over 100 students on MATLAB, Python, Excel to refine course concepts
- Mentor and guide teams of students through their projects, and navigation of their academic careers
- Run help sessions to provide personalized instruction to individual students and small groups
- Facilitate handling of supplies and materials for class and other organizational duties

Undergraduate Research Assistant at Ma Quantum Lab January 2021 – June 2021
Purdue University, West Lafayette IN

- Created and presented a poster using Latex at the Purdue Undergraduate Research Conference
- Researched Microwave electronics for superconducting quantum circuits for use on quantum simulation testing
- Realized noise reduction of 90% on a log curve for when trying to analyze signal received from system
- Collaborated with Principal Investigator and lab group to exchange idea for finding better methods for end-to-end testing of system

PROJECTS

Project ASTRA January 2022 – Present

- Utilized Kalman filtering and other filtration methods to predict and verify sensor data accuracy.
- Create and model motion of spacecraft using sensor data with MATLAB and Simulink
- Designed a rocket that can be controlled midair with PSP Active Controls group.
- Modeling 6-DOF and utilizing dead reckoning.

Filtering Analysis for GNSS Signal Correction August 2023 – December 2023

- Semester long research project in conjunction with Satellite Navigation Course.
- Utilizes a variety of different verified Filtering models for signal analysis of Satellite Navigation Systems.
- From independently sourced GNSS data, was able to get an accuracy of 5 meters and low mean square root error.
- Built mainly utilizing MATLAB and Python.
